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# Final Report:
# Adult Census Income Analysis

## Introduction and Problem Definition

This Capstone Project analyzes the UCI Adult Census Income dataset to determine which factors most strongly predict whether an individual earns more than $50,000 per year. The cleaned dataset contains 48842 rows and 15 columns, and after encoding and feature engineering the final modeling configuration uses 28 features. The client is a workforce development agency interested in understanding structural drivers of income inequality and designing interventions that improve economic mobility.

The analysis pipeline consists of data wrangling, exploratory data analysis, preprocessing and feature engineering, and supervised modeling. The objective is not only to accurately classify income category but also to interpret which demographic, educational, occupational, and financial factors contribute most to high-income outcomes in a way that is actionable for policy and program design.

## Data Wrangling

The data wrangling stage combines the original Adult Census training and test files into a single analytical dataset and applies a series of cleaning operations. Income labels from the raw files include formatting differences such as trailing punctuation, which are standardized into a clean binary label indicating whether income is above or at or below $50,000. Column names are aligned with the UCI documentation, and data types are enforced for numeric and categorical fields. Rows are retained rather than dropped, and there is no imputation because the cleaned dataset contains no ambiguous placeholder values. Summary checks confirm that numeric ranges and category sets are consistent with expectations for age, hours worked, capital gains, capital losses, workclass, occupation, marital status, relationship, race, sex, and native country.

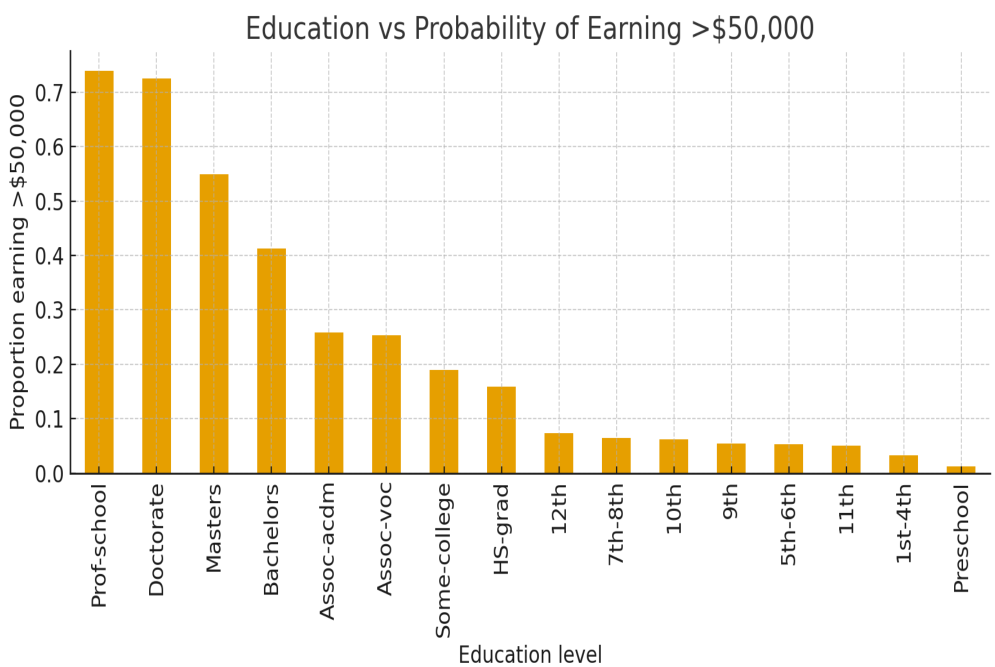
Additional integrity checks ensure that key distributions are sensible and that potential duplicate records do not introduce obvious bias. Duplicates are examined but not removed, as they represent legitimate combinations in the underlying population rather than clear data-entry errors. The result is a consistent, fully specified dataset that is ready for exploratory analysis, encoding, and modeling.

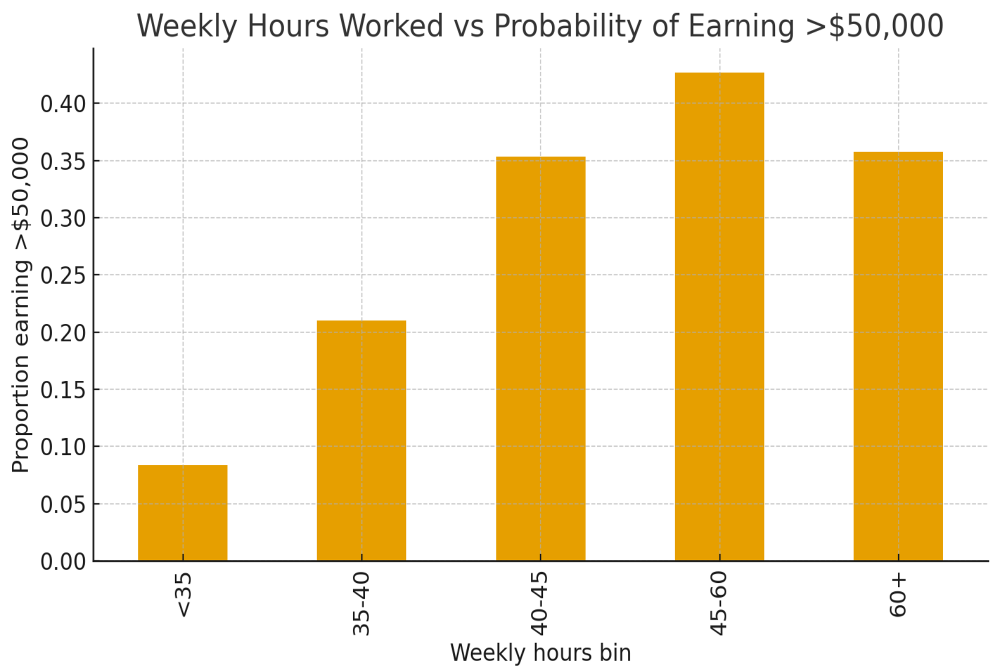
## Exploratory Data Analysis

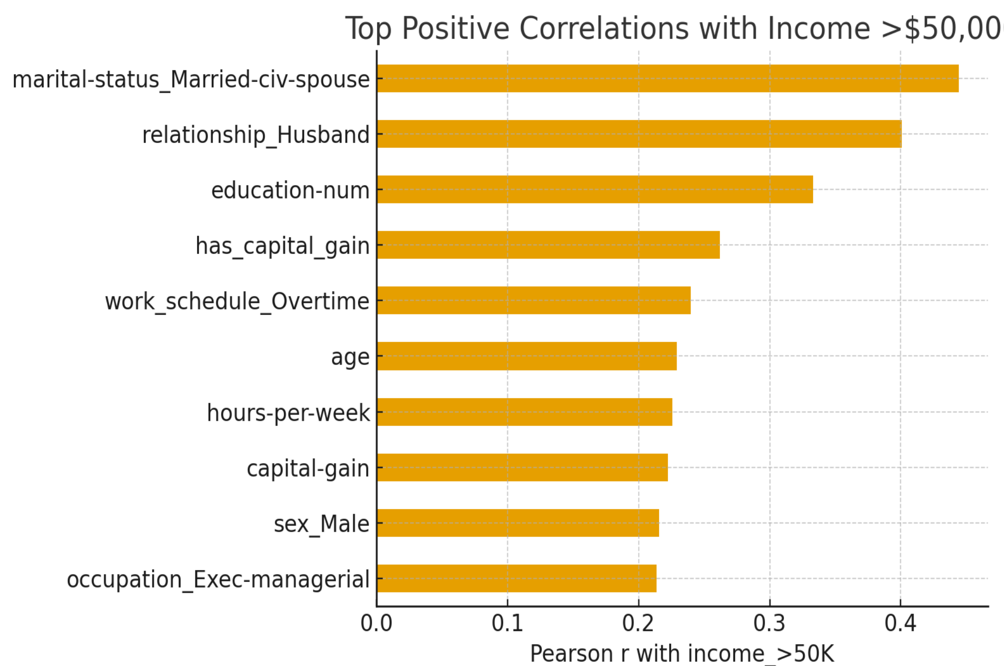
Exploratory analysis reveals strong relationships between education, weekly work hours, household structure, and income. Individuals with doctorate and professional degrees have probabilities above 0.70 of earning more than $50,000, whereas high school graduates remain below 0.15. The education-income visualization demonstrates a steep, monotonic increase in the proportion of high earners as education level rises, with particularly sharp gains at the bachelor, master, and professional tiers.

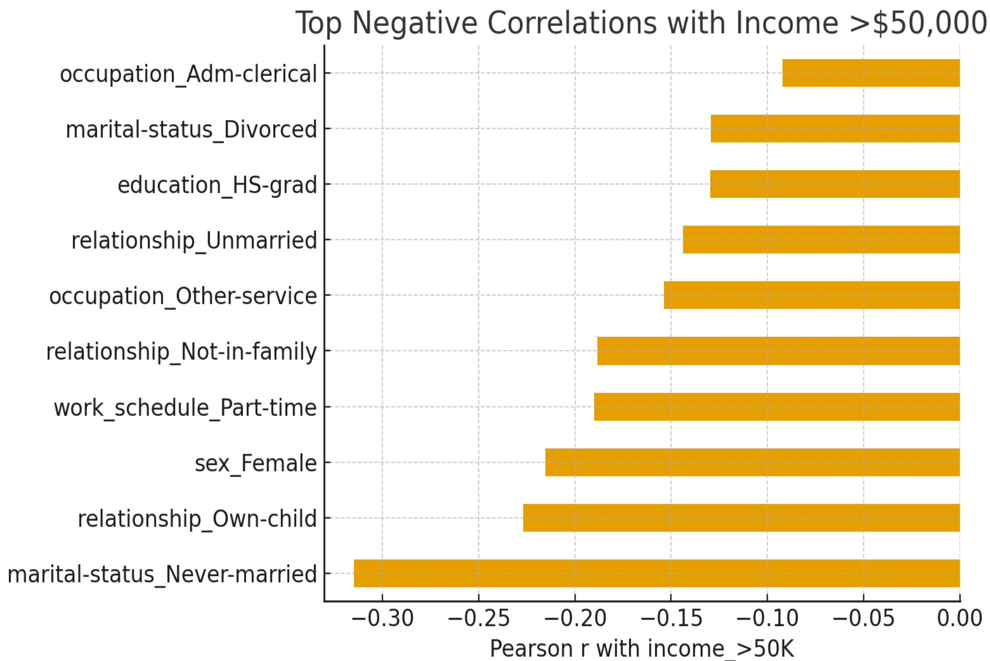
Weekly work hours follow a similar pattern. Individuals working fewer than 35 hours per week have probabilities below 0.10 of surpassing $50,000, while those in the 45–60 hour range exceed 0.40. This pattern indicates that higher income in this dataset is tightly coupled with full-time or extended work schedules, even though long hours alone are not sufficient to guarantee high earnings.

Pearson correlations computed on the encoded dataset provide a complementary, quantitative view of these relationships. Some of the strongest positive correlations with income_>50K are marital-status_Married-civ-spouse (r ≈ 0.44), relationship_Husband (r ≈ 0.40), education-num (r ≈ 0.33), and has_capital_gain (r ≈ 0.26). Strong negative correlations include marital-status_Never-married (r ≈ -0.31), relationship_Own-child (r ≈ -0.23), sex_Female (r ≈ -0.22), and work_schedule_Part-time (r ≈ -0.19). Together, these patterns show that education, household structure, gender, work schedule, and access to investment income are central to the income distribution captured in the data.




Weekly Hours Worked vs Probability of Earning >$50,000
Proportion earning >$50,000
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
<35
35-40
40-45
45-60
60+
Weekly hours bin


Top Positive Correlations with Income >$50,00
marital-status_Married-civ-spouse
relationship_Husband
education-num
has_capital_gain
work_schedule_Overtime
age
hours-per-week
capital-gain
sex_Male
occupation_Exec-managerial
0.0
0.1
0.2
0.3
0.4
Pearson r with income_>50K




## Preprocessing and Feature Engineering

Preprocessing transforms the 15 original columns into a richer set of encoded and engineered features. Categorical variables are one-hot encoded, and additional derived variables such as work schedule categories and capital-gain indicators are introduced. To reduce dimensionality where signal is clearly concentrated, some variables are deliberately simplified without comparison. In particular, race is binarized into an “is White” indicator, and native-country is binarized into an “is United States” indicator, reflecting their skewed distributions and the expectation that these simplified forms capture the relevant contrast for income modeling.

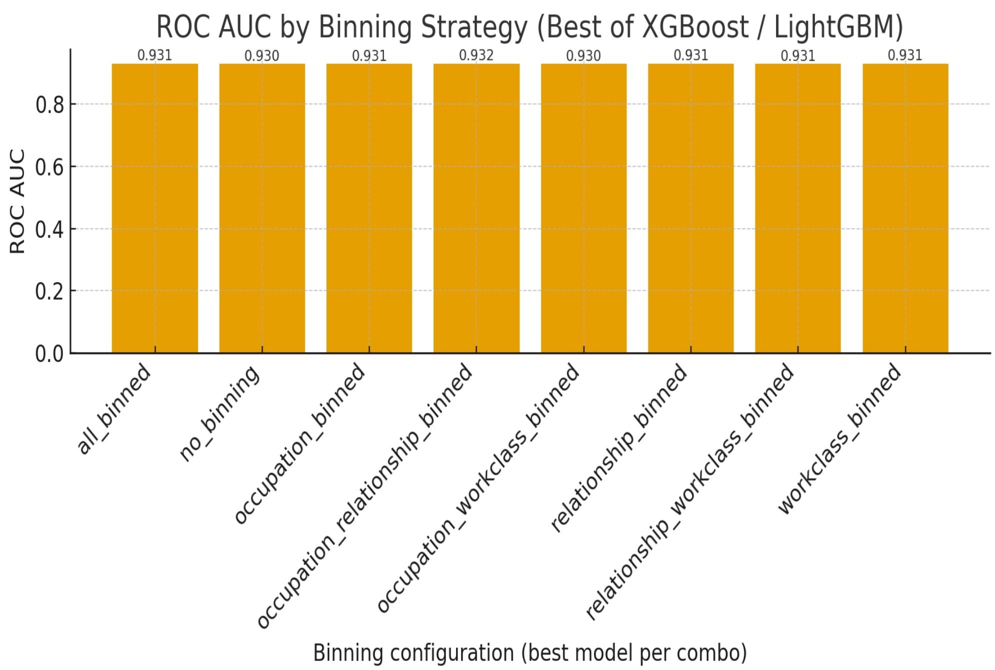
In contrast, three columns—relationship, occupation, and workclass—are treated more cautiously because it is not obvious a priori that binning will improve predictive performance. For these variables, the project constructs 8 feature configurations that represent all combinations of binned versus unbinned versions: no_binning, occupation_binned, relationship_binned, workclass_binned, occupation_relationship_binned, occupation_workclass_binned, relationship_workclass_binned, and all_binned. The all-binned configuration ultimately reduces the feature space to 28 columns while retaining the key socioeconomic distinctions that these variables encode. Each configuration is paired with stratified training and test splits so that model performance can be compared fairly across binning strategies.

## Model Performance and Interpretation

The modeling phase evaluates gradient-boosted tree models for each of the 8 binning configurations, using XGBoost and LightGBM with hyperparameters tuned via Optuna. For every configuration, Optuna explores candidate hyperparameter sets separately for XGBoost and LightGBM and records the best-performing model based on cross-validated ROC AUC. These best models are then retrained on the full training set for their respective configurations and evaluated on a held-out test set using multiple metrics, including ROC AUC, accuracy, precision, recall, F1-score, and average precision.

The evaluation summary shows that the strongest overall performer is a gradient-boosted XGB model trained on the occupation_relationship_binned configuration, with ROC AUC ≈ 0.932, accuracy ≈ 0.876, and precision ≈ 0.790. Across all 8 binning strategies, the best models per combo achieve ROC AUC values in a very narrow, high range, indicating that the boosted learners are consistently effective regardless of the exact grouping scheme. Models that include binned versions of relationship and occupation tend to rank near the top, suggesting that grouping categories along socioeconomic lines helps the model capture stable income patterns without sacrificing granularity where it matters most.

The bar chart of ROC AUC by binning configuration summarizes these comparisons visually. It shows that all 8 best-per-combo models perform at a similarly high level, with only modest differences between them. The exact ROC AUC for each configuration is displayed above its bar, making it clear that while occupation_relationship_binned with XGBoost is the top performer, configurations such as relationship_binned, workclass_binned, and all_binned are extremely close. This reinforces the conclusion that thoughtful feature engineering can simplify the representation of relationship, occupation, and workclass while preserving almost all of the predictive power that XGBoost and LightGBM are able to extract from the data.




## Conclusion

Overall, this project finds that income above $50,000 is strongly associated with higher educational attainment, longer and more intensive work schedules, married or partnered household structures, the presence of capital gains, and gender. Individuals who are never married, coded as own children in the household, female, or employed in part-time work schedules face systematically lower probabilities of earning above $50,000, even after accounting for other characteristics. The alignment between exploratory analysis, correlation statistics, and the performance of tuned XGBoost and LightGBM models provides a coherent narrative about how human capital, household structure, and labor patterns combine to shape income outcomes in this dataset.

## Future Research

Future extensions of this analysis could include fairness-aware modeling to quantify and mitigate disparate impact across gender, race, and household structure, as well as longitudinal studies to examine income mobility over time. Counterfactual simulations based on the trained boosted models could estimate how specific interventions—such as completing an additional degree, transitioning from part-time to full-time work, or changing occupation class—would affect the probability of crossing the $50,000 threshold for different demographic segments. These extensions would deepen both the methodological rigor and the policy relevance of the work by moving from descriptive and predictive insight toward explicit evaluation of intervention strategies.

## Recommendations

Based on these findings, a workforce development agency should prioritize expanding access to advanced education and occupation-specific training, particularly in fields associated with stable, higher earnings. It should also provide targeted support for unmarried individuals and single-parent households, for example through childcare, transportation, and scheduling assistance, to reduce barriers to full-time employment. Finally, the agency should work with employers and community partners to ensure equitable access to full-time and extended-hour opportunities so that women and other underrepresented groups are not disproportionately concentrated in part-time roles that structurally limit income potential.